

Open-Cell & Closed-Cell Laboratory Formulation Pack

Same 4-page lab dossier as the 29 Sep Rev. F file, with the pours and A-sides corrected. These are controlled R&D starting formulas, not a production specification or a fire certificate. TCPP lot data below is taken from the selected Elite Materials COA used in Rev. F.

1. Current status at a glance

System	Code	Status
Closed cell — pour first	BX-01	B-side defined. HFO-1233zd(E) 10.0 is in the 100. Index 125 → 50.00 g B + 41.40 g A . Not 1:1.
Open cell — pour if sucrose is on the floor	BX-02	B-side defined. Index 105 at sucrose OH 400 → 50.00 g B + 65.65 g A . Not 1:1.
Open cell — memo screen	OC-01	High-water System C. 12.00 g water + 16.50 g sucrose OH on 50 g B → 196.6 g A . A 1:1 shot is index ≈ 25 and is invalid.
Open cell — polyether screen from Rev. F	OC-F1	B-side only. Do not fix A at 50 g until Polyether 1, Polyether 2 and sucrose OH# are on the TDS.
Closed cell — older screen	CC-01	90.2 g A / 100 g B at index 120. If used, cut water to 1.4 and add 10.0 HFO, then re-index to 79.3 g A / 109.4 g B. Prefer BX-01.
TCPP	Selected lot	Elite Materials Corp. TCPP, lot XLT20260701. Non-reactive. OH contribution = 0.

2. What Rev. F got wrong

Point	Rev. F (other AI)	This pack
Closed-cell polyols	Stacked TB-307 + PE 104 + TEROL 250 + PE 103 as four polyesters	One high-OH drum (TB-307 or PE 104) + one flow drum (TEROL 250 or PE 103) + PD 230 trim. PE 104 is not a second backbone beside TB-307.
Closed-cell mix	Forced 50 g A : 50 g B by raising water to 2.09 so index hits 120 at 1:1	Set water for blow (1.20 with HFO 10.0), then pay NCO. 50 g B takes 41.40 g A at index 125.
Closed-cell catalysts	K-15 / DMCHA / K-KAT / bismuth	PO 1009433 pack: PC-15, DABCO 2040, CAT-T, PC-37.
Open-cell A-side	Wanted 1:1 later; A left blank	Pay water NCO first, then polyol OH. BX-02 = 65.65 g A. OC-01 = 196.6 g A. Never pour 20–28 g A on the memo cup.

3. TCPP — selected lot (from Rev. F COA)

Chemical	Tris(2-chloroisopropyl) phosphate	APHA / viscosity @25 °C	10 · 66 cP
CAS	13674-84-5	Phosphorus / chlorine	9.4 wt% · 32.4 wt%
Lot	XLT20260701 · Elite Materials Corp.	SG @20 °C / RI	1.29 · 1.46
Appearance	Colourless transparent liquid	Cost input on COA	USD 0.43/lb ≈ USD 0.948/kg
Water / acid no.	0.026 wt% · 0.015 mg KOH/g	Index treatment	OH contribution = 0

4. A-side method — water NCO, then polyol OH

$EW_{\text{polyol}} = 56\,100 / \text{OH\#} \cdot EW_{\text{water}} = 9.0 \cdot EW_{\text{PMDI}} = 4\,200 / \% \text{NCO} = \mathbf{135.5}$ at 31.0 % NCO.

A-side (g) = OH eq × (Index / 100) × 135.5. Do not force 1:1 and then back-calculate water.

Card	Water → PMDI at index 100	Polyol OH → PMDI at index 100	Index	A / 100 g B	50 g B cup
BX-01	1.20 → 18.07 g	TB-307 + TEROL + PD 230 → 48.17 g	125	82.8 g	41.40 g A
BX-02	5.20 → 78.29 g	Sucrose OH 400 + TEROL → 46.76 g	105	131.3 g	65.65 g A
OC-01	24.0 → 361.3 g	Sucrose 33.0 OH 400 → 31.9 g	100	393 g	196.6 g A
CC-01	2.00 → 30.11 g	TB-307 + TEROL + PD 230 → 45.03 g	120	90.2 g	screen only
CC-01 + HFO	1.40 → 21.08 g	Same polyols → 45.03 g	120	79.3 g on 109.4 g B	prefer BX-01

5. OPEN CELL — BX-02 preferred **POUR THIS**

Commercial-density open cell when sucrose is on the floor. No HFO. No polyether 1 / 2. TCPP lot is not in this card — TEP + reactive FR do the fire work. Sucrose OH taken as 400 until the lot TDS is in; then re-index.

Ingredient	Role	pbw	g / 50 g B	OH#	eq / 100 g
Sucrose / sucrose polyol	Network that can open and still stand	38.0	19.00	400	0.2709
TEROL 250 / KIMpol PE 103	Aromatic fire + flow. One drum, not both.	16.0	8.00	260	0.0742
TEP	Viscosity + phosphorus	20.9	10.45	0	0
Reactive FR	Bound FR. Re-index if the grade has OH.	7.0	3.50	0*	0*
Ecosurf SA-9	Emulsifies 5.2 water	4.0	2.00	0	0
Open-cell silicone	Opens windows	1.5	0.75	0	0
Water	Main chemical blow. Between memo 24 and Rev. F 4.5	5.2	2.60	6233	0.5778
PC-37	Blow / cream	1.70	0.85	—	—
CAT-T	Blow / cream	1.60	0.80	—	—
PC-15	Gel. Watch for closed windows	2.80	1.40	—	—
BDMAEE or DEDEE	Window rupture	0.90	0.45	—	—
Biocide	In-can protection	0.40	0.20	—	—
B-side total		100.00	50.00		0.9229
A-side PMDI 31.0 % NCO	Index 105	131.3	65.65		

Water 2.60 g ÷ 9.0 = 0.2889 eq → **41.10 g** PMDI at index 105. Sucrose 19.00 (OH 400) + TEROL 8.00 (OH 260) → **24.55 g** PMDI at index 105. Cup = **50.00 g B + 65.65 g A**. If sucrose OH is 350, A ≈ 62 g. If 450, A ≈ 70 g.

6. OPEN CELL — OC-01 memo screen (do not pour 1:1)

Ingredient	pbw	g / 50 g B	Ingredient	pbw	g / 50 g B
Sucrose / sucrose polyol	33.0	16.50	Reactive FR	8.0	4.00
TEP	17.0	8.50	Ecosurf SA-9	6.0	3.00
Open-cell silicone	1.2	0.60	Water	24.0	12.00
PC-37	2.4	1.20	CAT-T	2.4	1.20
PC-15	4.2	2.10	BDMAEE / biocide	1.0 / 0.8	0.50 / 0.40
B-side	100.0	50.00	A-side, index 100		196.6

Water 12.00 ÷ 9.0 = 1.3333 eq → 180.67 g PMDI. Sucrose 16.50 · OH 400 · EW 140.25 → 15.94 g PMDI. Total **196.6 g A**. The old 20–28 g A charge never paid the water.

7. OPEN CELL — screening alternatives (change one variable)

Code	Change from BX-02	Purpose
BX-02	Control · 50.00 g B + 65.65 g A	Baseline 16–22 kg/m³
BX-02a	Water 5.20 → 4.80; TEP +0.40	Higher density / more stable rise
BX-02b	Water 5.20 → 5.60; TEP –0.40 · re-index A (+6.0 g / 100 g B)	Lower density / more opening
BX-02c	PC-15 2.80 → 2.60	If windows stay too tight
BX-02d	PC-37 + CAT-T each +0.15	If cream is slower than 4 s
OC-01	Memo screen · 196.6 g A on 50 g B	Only after BX-02; do not blend the two B-sides

8. OPEN CELL — initial target window

Measurement	BX-02	OC-01 screen
A:B by weight	65.65 : 50.00 · not 1:1	196.6 : 50.00 · not 1:1
Cream / rise / tack-free	2–4 s / record / record	2–5 s / 8–15 s / 10–20 s
Free-rise density	16–22 kg/m³	7–12 kg/m³
Open-cell content	> 95 %	< 10 % closed
Collapse	None. If collapse: +0.15 g PC-15 on the 50 g cup	None

10. CLOSED CELL — BX-01 preferred **POUR FIRST**

Use one high-OH polyester (ISOEXTER TB-307 or KIMpol PE 104, midpoint OH 305, f = 2.5) and one flow polyester (TEROL 250 or KIMpol PE 103, midpoint OH 260). Do not charge all four drums into one cup. HFO-1233zd(E) is mandatory. Water is 1.20 because HFO does the blow — do not raise water to force 1:1.

#	Ingredient	Role	pbw	g / 50 g B	OH#	eq / 100 g
1	TB-307 / PE 104 — write which	Rigid network. Replaces Mannich / CREAPOL 326	41.3	20.65	305	0.2245
2	TEROL 250 / PE 103 — write which	Flow + aromatic fire. Replaces CREAPOL 1500	24.8	12.40	260	0.1149
3	CREAPOL PD 230	Toughness trim. No substitute on the floor	4.0	2.00	225	0.0160
4	TCPPI lot XLT20260701	Halogen FR · 0 OH. Terol already does part of the fire job	6.0	3.00	0	0
5	TEP	Viscosity cut for thick TB-307 / PE 104	4.0	2.00	0	0
6	Reactive FR	Bound phosphorus. Re-index if it has OH	2.0	1.00	0*	0*
7	Emulsifier	Holds 1.20 water in the polyester	1.75	0.875	0	0
8	Closed-cell silicone	Sealed cells. Not the open-cell grade	1.5	0.75	0	0
9	Water	Nucleation only. Do not go back to 1.75 or 2.09	1.20	0.60	6233	0.1333
10	PC-15	Gel	1.45	0.725	—	—
11	DABCO 2040	Gel / cure	1.10	0.550	—	—
12	CAT-T	Cream / blow	0.55	0.275	—	—
13	PC-37	Cream / blow	0.35	0.175	—	—
14	HFO-1233zd(E)	Mandatory blowing gas. Last, 15–18 °C, cap	10.0	5.00	0	0
	B-side total		100.00	50.000		0.4888
A	PMDI 31.0 % NCO	Index 125	82.8	41.40		

Water 0.60 g + 9.0 → **11.29 g** PMDI at index 125. Polyols 20.65 + 12.40 + 2.00 → **30.11 g** PMDI at index 125. Cup = **50.00 g B + 41.40 g A**. If the flow drum is TEROL 250 at EW 224.4, A is about 41.05 g. Hood on. Mix 4–5 s.

Rev. F CC-F1 (TB-307 40.06 + PE 104 18.03 + TEROL 10.01 + PE 103 5.01 + water 2.09 + HFO 9.5 + 50 g A) is withdrawn. The 1:1 index was obtained by stacking equivalent drums and adding water. That is not the catalyst-package method.

11. CLOSED CELL — screening alternatives (from BX-01, not from Rev. F CC-F2/F3)

Code	Change from BX-01	Purpose
BX-01	Control · 50.00 g B (5.00 g HFO in) + 41.40 g A	Does it seal at 32–38 kg/m³?
BX-01a	HFO 10.0 → 11.0 / water 1.20 → 1.10 · re-index A slightly down	Lighter / more physical blow
BX-01b	CAT-T +0.15 / PC-15 +0.15	If cream > 6 s or gel > 15 s
BX-01c	CAT-T -0.10 / PC-15 -0.10	If it scorches or snaps too fast
CC-01	Older 90:100 screen + HFO 10 / water 1.4 → 79.3 g A / 109.4 g B	Fallback only if BX-01 is too tight

12. CLOSED CELL — initial target window

A:B by weight	41.40 : 50.00 · ≈ 83 : 100	Closed-cell content	> 90 %
Nominal NCO index	125 (calc.)	24 h shrinkage	No visible shrink / distortion
Cream / gel / tack-free	3–6 s / 8–15 s / 15–28 s	Cell structure	Fine, sealed, uniform
Free-rise density	32–38 kg/m³	HFO handling	B cup 15–18 °C · weigh last · cap

13. Recommended first cup-test sequence

Order	Trial	Cup	Purpose
1	BX-01	50.00 g B + 41.40 g A	Preferred closed-cell control. HFO in.
2	BX-01 repeat ×3	Same	Repeatability before any catalyst change.
3	BX-02	50.00 g B + 65.65 g A	Preferred open cell if sucrose is on the floor. No HFO.
4	OC-01	50.00 g B + 196.6 g A	Memo screen only. Do not blend with BX-02.
5	OC-F1	50.00 g B only	Wait for the three polyol OH# values before weighing A.

14. Cup-test method

Step	Method
1	Condition A at 21–23 °C. Closed-cell B is chilled to 15–18 °C after HFO. Open-cell A and B stay 21–23 °C. Record actual temperatures.
2	Dry cups. Weigh to 0.01 g. Prepare B exactly to 50.00 g in the order on the card. Water before catalysts.
3	Homogenize B. For BX-01: HFO last, cap at once, hood on. If HFO flashes off, recap and reweigh — do not add water to compensate.
4	Weigh A separately: 41.40 g (BX-01), 65.65 g (BX-02), or 196.6 g (OC-01). Do not default to 50 g A.
5	Pour A into B. Mix 4–5 s at 2 500–3 000 rpm. Same mixer and time on every comparison cup.
6	Clock cream, gel/string, rise and tack-free from mix start.
7	Do not disturb after cream. Cut and inspect at 24 h.
8	Record free-rise density, cell structure, collapse, shrinkage, friability and scorch.

15. Trial record sheet

Field	Result	Field	Result
Trial code		Date / operator	
B-side mass	50.00 g	A-side mass (write)	
A:B ratio		Calculated index	
High-OH drum	TB-307 / PE 104	Flow drum	TEROL 250 / PE 103
Sucrose grade / OH#		PMDI lot / %NCO	
A / B temperature		Mix rpm / time	
Cream	s	Gel / string	s
Rise	s	Tack-free	s
Free-rise density	kg/m ³	Open / closed observation	
24 h shrinkage		Collapse / scorch / friability	
Comments			

16. OPEN CELL — OC-F1 polyether screen (Rev. F B-side kept; A-side not fixed)

Water 4.5 already needs 33.9 g PMDI at index 100 before any polyol OH. Do not weigh 50 g A until the three OH# values arrive.

Ingredient	pbw	g / 50 g B	Ingredient	pbw	g / 50 g B
Polyether polyol 1	24.0	12.00	Polyether polyol 2	22.0	11.00
Sucrose / polyether polyol	15.0	7.50	CREAPOL 1500	5.0	2.50
TCPPO lot XLT20260701	23.0	11.50	Water	4.5	2.25
Blow pack PC-37 / CAT-T / BDMAEE	1.5	0.75	PC-15	1.0	0.50
DABCO 2040	0.5	0.25	Emulsifier / opener	2.0	1.00
Open-cell silicone	1.5	0.75	B-side · A not fixed	100.0	50.00

17. Critical inputs still to confirm

Input	Why it matters
Which high-OH drum is open	TB-307 and PE 104 are alternatives, not a stacked pair. Write the label before the first BX-01 cup.
Which flow drum is open	TEROL 250 vs PE 103. If the drum is TEROL 250 at EW 224.4, BX-01 A is about 41.05 g on 50 g B.
Sucrose / sucrose-polyol OH#	BX-02 and OC-01 A-sides move if the lot is not 400. Water term does not change.
OC-F1 Polyether 1 / 2 OH#	Required before any A-side is assigned to the Rev. F polyether screen.
Actual PMDI %NCO from COA	Recalculate EW = 4 200 / %NCO. Do not keep 135.5 if the lot is not 31.0 %.
Exact silicone grades	Open-cell vs closed-cell. Do not swap them.
Reactive FR OH	Include in the index if the selected grade carries OH.

Change one variable at a time. If HFO or water changes, keep the B-side intent clear and re-index A — do not hold 50 g A. High-water open cell is never 1:1. Closed-cell cups without HFO are not valid on this pack.

Prepared 2 Oct 2026 from the BazzLab pack, the catalyst-package water-then-polyol method, PO 1009433 cats, and the TCPPO COA used in Rev. F. The website pack is Rev. M. ISOEXTER, TEROL, KIMpol, CREAPOL, DABCO, Honeywell and Elite Materials names remain the property of their owners.